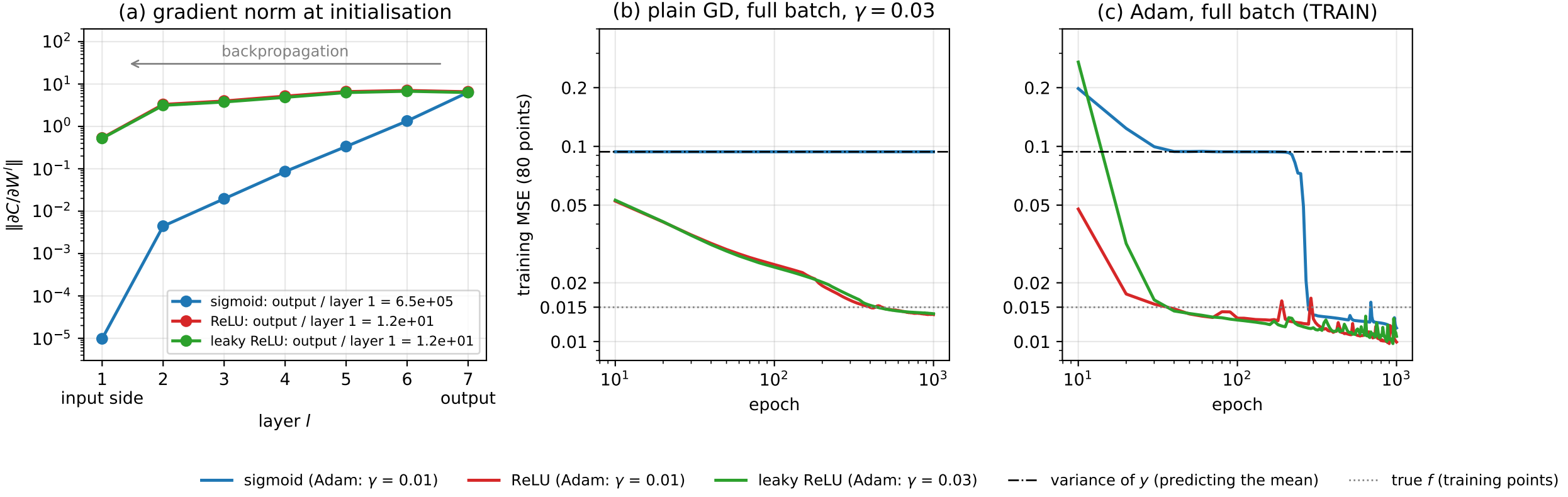
(a) gradient norm at initialisation
backpropagation
$\|\partial C/\partial W^l\|$
sigmoid: output / layer 1 = 6.5e+05
ReLU: output / layer 1 = 1.2e+01
leaky ReLU: output / layer 1 = 1.2e+01
input side
output
layer l
(b) plain GD, full batch, $\gamma = 0.03$
training MSE (80 points)
epoch
(c) Adam, full batch (TRAIN)
epoch
sigmoid (Adam: γ = 0.01)
ReLU (Adam: γ = 0.01)
leaky ReLU (Adam: γ = 0.03)
variance of y (predicting the mean)
true f (training points)